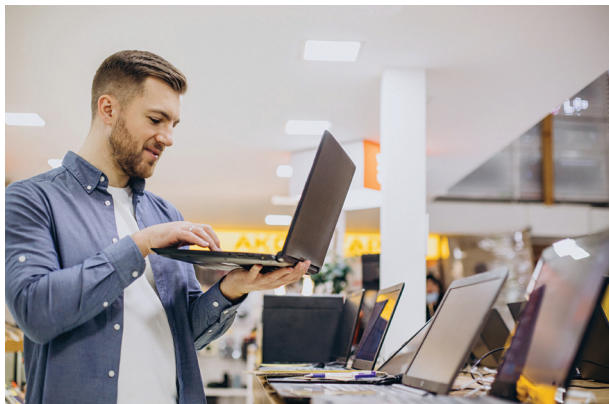


Overview of tech waste and pollution

- **E-Waste Accumulation:** Electronic waste (e-waste) is one of the fastest-growing waste streams globally. Discarded devices often end up in landfills, where toxic substances like lead, mercury, and cadmium can leach into the soil and water.
- **Resource Depletion:** The manufacturing of gadgets requires precious metals and rare earth elements. Mining these resources leads to habitat destruction and pollution.
- **Energy Consumption:** Producing and powering devices consumes significant amounts of energy, much of which comes from non-renewable sources, contributing to carbon emissions.

Carbon footprint of popular devices

- **Smartphones:** The average smartphone contributes approximately 80 kg of CO2 emissions over its lifecycle, with most emissions occurring during production.
- **Laptops and Tablets:** Larger devices typically have a higher carbon footprint due to more substantial material usage and energy consumption during manufacturing.



- **Data Centers:** The increased demand for cloud services requires vast data centres, which consume large amounts of electricity for operation and cooling.

How to select sustainable tech brands

Choosing products from companies committed to environmental stewardship can make a significant difference.

How to identify eco-friendly tech companies

- **Environmental Policies:** Research a company's commitment to sustainability on their official website or sustainability reports.
- **Transparency:** Companies that openly share information about their supply chains, manufacturing processes, and environmental impact are often more trustworthy.
- **Sustainable Practices:**
 - **Use of Recycled Materials:** Some companies incorporate recycled plastics and metals into their products.

- **Energy Efficiency:** Devices designed to consume less power during operation.

Certifications and indicators of sustainability

- **Energy Star Certification:** Indicates that a product meets energy efficiency guidelines set by the U.S. Environmental Protection Agency.
- **EPEAT Rating:** The Electronic Product Environmental Assessment Tool rates products based on environmental criteria, including material selection, design for end-of-life, and corporate performance.
- **Fair Trade and Ethical Sourcing:** Certifications that ensure materials are sourced responsibly, and workers are treated fairly.

Buying refurbished or second-hand devices

Opting for refurbished or pre-owned gadgets extends the life of products and reduces demand for new device manufacturing.

Where to buy refurbished gadgets

- **Manufacturer Programs:** Companies like Apple, Dell, and Samsung offer certified refurbished products with warranties.
- **Trusted Retailers:** Websites like Amazon Renewed, Best Buy, and Newegg provide refurbished electronics with quality assurances.
- **Certified Refurbishers:** Look for sellers who are authorized or certified refurbishers to ensure the device has been properly restored.

Key checks before purchasing refurbished tech

- **Warranty Coverage:** Ensure the device comes with a warranty to protect against defects.
- **Return Policy:** Verify the return policy in case the device doesn't meet expectations.
- **Device Condition:** Check for descriptions like "Grade A" or "Like New" for minimal wear and tear.
- **Reviews and Ratings:** Read customer feedback about the seller and product quality.

2) Recycling and upcycling old devices

Properly disposing of or repurposing old electronics prevents environmental harm and can give devices a second life.

Safe disposal and recycling options

Electronics contain hazardous materials that require careful handling when disposed of.

Check for electronics recycling programs

- **Manufacturer Take-Back Programs:** Companies like Apple, HP, and Samsung offer programs where you can return old devices for recycling.
- **Retailer Programs:** Stores such as Best Buy and Staples